

Activity 1: The Shrinking Secant

IBDP Mathematics AA HL — Polynomial Functions & The Intuitive Foundation of Limits

Name: _____ Date: _____

Instructions

- 1. Complete all parts in the spaces provided. Show all working clearly.
- 2. Use a calculator for numerical exploration but leave algebraic steps exact.
- 3. Read each TOK reflection box and write a short response (2–3 sentences).
- 4. The final bridge question connects to Activity 2—keep your answer for reference.
- 5. Marks are indicated in square brackets. This activity should take approximately 120 minutes.

Section A: Numerical Exploration — The Secant That Shrinks [9 marks]

- 1.1** Consider $f(x) = (1/2)x^2 + 2$. We wish to find the exact gradient of the curve at $x = 2$. [3]
 (a) Write the slope of the secant line passing through $(2, f(2))$ and $(2 + h, f(2 + h))$ as a single algebraic expression in h . Simplify fully. [3 marks]
- 1.2** Build a table for $h = 1, 0.1, 0.01, 0.001, -0.001, -0.01, -0.1, -1$, calculating the secant slope to 6 decimal places. What value does the slope appear to be approaching? [3 marks]
- 1.3** Explain why we must examine both $h \rightarrow 0^+$ and $h \rightarrow 0^-$ rather than only positive h . What geometric concept is lost if we only consider one side? [3 marks]

Section B: Algebraic Formalisation — The Limit of the Difference Quotient [7 marks]

- 2.1** Evaluate $\lim_{h \rightarrow 0} [f(2+h) - f(2)] / h$ algebraically using your simplified expression from Section A. Show every algebraic step. [3 marks]
- 2.2** Generalize: For $f(x) = (1/2)x^2 + 2$, find the derivative function $f'(x)$ at any point $x = a$ by evaluating $\lim_{h \rightarrow 0} [f(a+h) - f(a)] / h$. Show your working. [4 marks]

Section C: Proof — The Power Rule from First Principles [6 marks]

- 3.1** Let $f(x) = x^3$. Using the difference quotient and the binomial expansion $(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$, prove that $f'(x) = 3x^2$ directly from the limit definition. State each algebraic step clearly. [4 marks]
- 3.2** Justify: Explain why, after expanding $(x+h)^n$ for any positive integer n , the limit as $h \rightarrow 0$ always yields nx^{n-1} . You do not need to write the full binomial expansion—explain the reasoning. [2 marks]

Section D: Continuity & the IVT — Root Existence [3 marks]

- 4.1** Let $p(x) = 2x^3 - 7x^2 + x + 10$. (a) Evaluate $p(-1)$ and $p(0)$. (b) State the Intermediate Value Theorem. (c) Deduce that $p(x)$ has at least one root in $(-1, 0)$, justifying each condition of the IVT holds. [3 marks] [3]

TOK Reflection

- 5.1** The Concept of Infinity: Calculus relies on the infinitely small ($h \rightarrow 0$). Can the human mind truly grasp the 'infinitely small,' or is this just a useful linguistic trick we invented to make our formulas work? Write 2-3 sentences. [0]

Bridge to Activity 2

- 6.1** In Section B you cancelled the factor h from numerator and denominator. In Activity 2 you will encounter functions where the problematic factor cannot be cancelled so simply. Write down one example of a rational function that would be undefined at $x = 2$ and predict whether the limit exists. [0]